

TECHARENA

Global Talent Building the Future Together

Italy

AI cuts RF tolerance analysis from 277 days to 15 hours

Sub-topic 3 | A physics-checked AI model that stands in for the full-wave solver

A tolerance study takes months to a century of solver time

30 min

one full-wave simulation

per board design

277 days

sensitivity analysis

solver only: 13,312 simulations

110 years

design search

solver only: 1.92 M simulations

30%

of boards pass today

0.3 dB margin vs 1.3 dB factory spread

Serial solver time at 30 minutes per simulation. Pass rate under illustrative acceptance limits.

Finding and fixing the causes with the solver alone is not affordable.

One AI model, trained on 60 simulations, replaces the solver

- Learns how the 11 manufacturing tolerances shape the S-parameters.
- Obeys the physics: reciprocity built in, passivity checked on every prediction.
- 60 simulations are enough: ten random 60-simulation models match the full-data results within about 3 points.

How it works

60 full-wave simulations

11 manufacturing tolerances in, S-parameters out



AI model (kriging)

predicts every S-parameter in milliseconds

physics: reciprocity built in, passivity checked



Sensitivity analysis

which tolerances matter

KPI 2



Design search

which design passes most

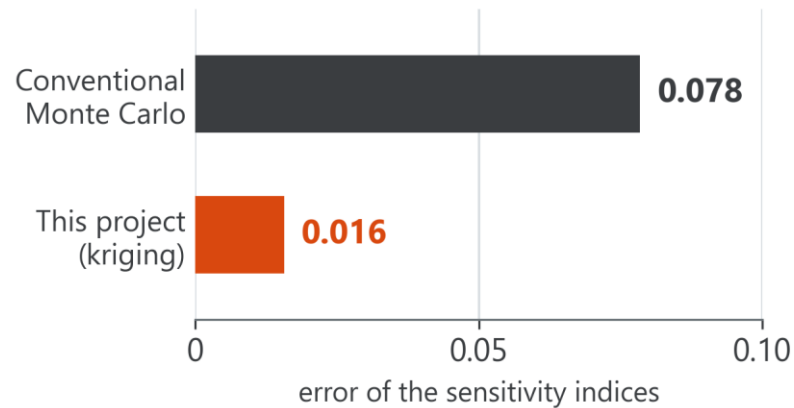
KPI 1

Thirty simulations find the tolerances that matter, with 5× lower error

Sensitivity analysis (KPI 2)

both from the same 30 simulations; lower is better

5x lower error



Polynomial chaos and a neural network reach the same accuracy as kriging.

Which tolerances matter, from 45 simulations

circle and line: estimate with 90% bootstrap interval
diamond: the same analysis on all 540 runs

Reflection

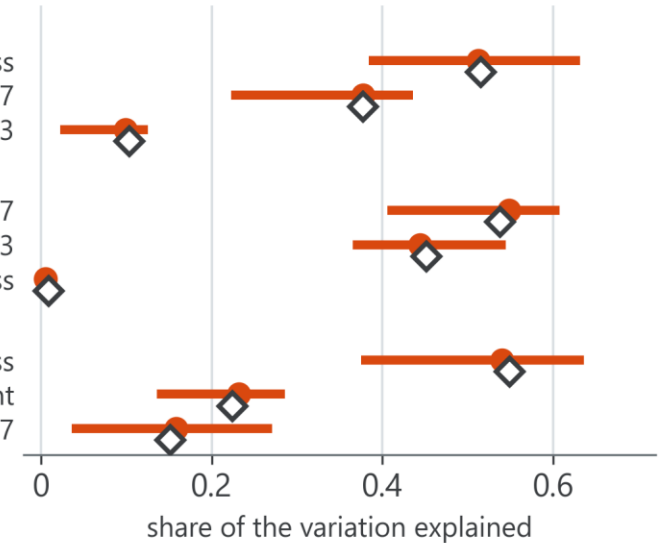
board thickness
copper width A7
copper width A3

Coupling

copper width A7
copper width A3
board thickness

Isolation

board thickness
material constant
copper width A7



three leading drivers per metric, Huawei tolerances, uniform scatter

Board thickness and copper widths A7 and A3 drive most of the variation: the tolerances worth tightening.

A 110-year design search now takes 30 hours

Design search (KPI 1): 961 candidate designs, each tested on 2,000 virtual boards

Solver only

110 years

1.92 million simulations, one after another

This project

30 hours

60 simulations to train, then 5 minutes of computing

Serial time at 30 minutes per simulation. With 100 solver licences in parallel, the solver-only search still takes about a year.

The same 60-simulation model also delivers the sensitivity analysis: no extra simulations needed.

Four design changes lift the pass rate to 73–85%

Changes: higher laminate constant; tighter A7, board-thickness and A3 tolerances.

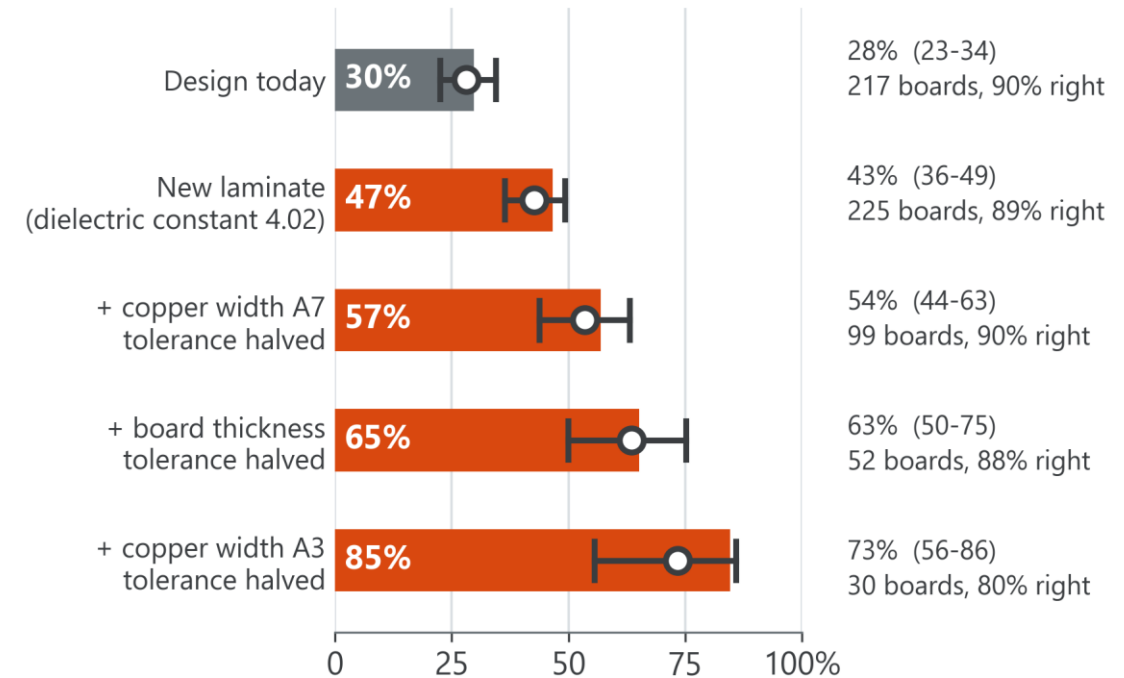
Each prediction was made without that scenario's real boards: 5 of 5 fall within the real range.

Holds when the illustrative limits are halved or tripled (92–97% agreement).

0 of 734 million predictions break the physics (passivity).

Boards passing under the stated limits

bar = model prediction, made without that scenario's real boards
circle and line = the real boards, with their 95% range



each design change keeps the ones above it

last step: 30 real boards; the 11-point gap is within sampling noise

Established methods, applied with physics checks and real-board validation

	Common practice in the literature	This project
Surrogate	Kriging / Gaussian process, about a third of RF surrogate studies [4]	Kriging, checked against polynomial chaos and a neural network on the same boards
Output	Compress the frequency response first (PCA or response features)	No compression: PCA raised the error 2.8x on this data, so all 3,012 outputs are predicted
Physics	Physics-informed networks put Maxwell's equations in the loss; they need field data [6]	Only port data exists: reciprocity built in, passivity checked on 734 million matrices
Sensitivity	Closed-form Sobol indices from a GP [1-3]; kriging-PCE in EM dosimetry [5]	Applied to the 11 manufacturing tolerances; leading tolerances found from 30 simulations
Pass rate	GP-assisted Monte Carlo that re-checks borderline boards in the solver [7]	No extra simulations allowed: every pass rate checked against held-out real boards

[1] Oakley & O'Hagan (2004), J. R. Stat. Soc. B 66(3)

[2] Marrel et al. (2009), Reliab. Eng. Syst. Saf. 94(3)

[3] Le Gratiet, Cannamela & Iooss (2014), SIAM/ASA J. Uncertain. Quantif.

[4] Systematic review, surrogate-based EM design of RF components (2026), Sensors 26(8):2504

[5] Kersaudy et al. (2015), J. Comput. Phys. 286:103-117

[6] Khan & Lowther (2022), IEEE Trans. Magn. 58(9)

[7] Fuhrländer & Schöps (2020), J. Math. Ind. 10

Project folder

Scan to open the full material

README: pipeline, evidence and limitations

Code that reproduces every number

Stored results of every study

github.com/BuseDuygun22/rf-tolerance-surrogates





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THANK YOU

From 277 days of solver time to 15 hours – with the physics checked on every prediction.

60 simulations · exact sensitivity analysis · pass rates checked on held-out real boards